Polymer in theta solvent

Complex disordered systems: Polymers

Consider a polymeric molecule in a solvent at temperature $T_1 = \Theta$ (i.e. at the so-called *theta temperature* of the solvent). It is composed of $N = 10^5$ identical monomers of size $b = 0.2$nm.

1. Estimate its root-mean-square end-to-end distance.
2. Provide a second estimate at a second temperature $T_2 \gg \Theta$

Solution

The problem simply requires us to remember the following:

1. The temperature Θ is the one at which the freely-jointed chain model is most suitable. In such case $R_{rms} = \sqrt{\langle R^2 \rangle} = bN^{1/2} \approx 63$nm
2. For temperatures much higher than Θ the chain is swollen and excluded-volume effects dominate, so that $R_{rms} = \sqrt{\langle R^2 \rangle} = bN^{3/5} = 200$nm